

Faisal Ahmed Sifat

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Research Interests

Distributed Machine Learning · Federated and Decentralized Learning · AI Systems · Edge and Cloud Computing
· Resource-Efficient AI

Publications

Md Sahadul Hasan Arian*, **Faisal Ahmed Sifat***, Saif Ahmed, Nabeel Mohammed, and Taseef Hasan Farook. "Unsupervised tooth segmentation from three dimensional scans of the dental arch using domain adaptation of synthetic data." *International Journal of Medical Informatics*, vol. 195, p. 105769, 2025. (**Q1, IF 3.2**)

* Equal contribution

Taseef Hasan Farook, Saif Ahmed, Farah Rashid, **Faisal Ahmed Sifat**, Preena Sidhu, Pravinkumar Patil, Sumaiya Zabin Eusufzai, Nafij Bin Jamayet, James Dudley, and Umer Daoood. "Application of 3D neural networks and explainable AI to classify ICDAS detection system on mandibular molars." *The Journal of Prosthetic Dentistry*, vol. 133, no. 5, pp. 1333–1341, 2025. (**Q1, IF 4.1**)

Faisal Ahmed Sifat*, Md Sahadul Hasan Arian*, Saif Ahmed, Taseef Hasan Farook, Nabeel Mohammed, and James Dudley. "An application of 3D vision transformers and explainable AI in prosthetic dentistry." *Applied AI Letters*, vol. 5, no. 4, p. e101, 2024.

* Equal contribution

Md Sahadul Hasan Arian*, **Faisal Ahmed Sifat***, Saif Ahmed, Nabeel Mohammed, Taseef Hasan Farook, and James Dudley. "Dental Loop Chatbot: A prototype large language model framework for dentistry." *Software*, vol. 3, no. 4, pp. 587–594, 2024.

* Equal contribution

Research Experience

AI/ML Researcher

Jul. 2024 – Jan. 2025

North South University & University of Adelaide Collaboration

- Implemented a PointNet++ domain adaptation pipeline for unsupervised segmentation of 3D intraoral CBCT scans using synthetically generated dental arch data; published in *International Journal of Medical Informatics* (Q1, IF 3.2).
- Developed Vision Transformer (ViT) models operating on 3D point cloud representations of dental prosthetics, incorporating Grad-CAM explainability; published in *Applied AI Letters*.
- Designed and evaluated retrieval-augmented generation (RAG) pipelines for domain-specific dental information retrieval, contributing to an LLM-based clinical decision support prototype; published in *Software*.
- Applied 3D convolutional neural networks with SHAP-based explainability for caries classification on mandibular molar scans under the ICDAS detection system; published in *Journal of Prosthetic Dentistry* (Q1, IF 4.1).
- Collaborated with researchers at the University of Adelaide across all four publications.

Education

B.Sc. in Computer Science and Engineering

Jan. 2020 – Nov. 2023

North South University, Dhaka, Bangladesh

CGPA: 3.18 / 4.00

- Relevant Coursework:** Pattern Recognition, Machine Learning, Operating Systems, Database Systems, Probability & Statistics, Algorithms, Software Quality Assurance.
- Thesis / Capstone:** Applied deep learning methods for medical image analysis; supervised by Dr. Nabeel Mohammed.

Professional Experience

Software Engineer (Contract)

Feb. 2026 – Present

All Generation Tech, Remote

- Developing enterprise-grade dashboards and workflow systems for US-based Enterprise clients.

Founder & Software Engineer

Mar. 2024 – Present

AlphaWolf Ventures, Remote

- Designed a fault-tolerant distributed workflow execution engine using Redis Streams, BullMQ, and Kubernetes, supporting parallel task scheduling and isolated execution of AI workloads across heterogeneous compute nodes.
- Implemented a CQRS-based event-driven microservice architecture for a multi-tenant platform, handling asynchronous job dispatch, worker coordination, and failure recovery at scale.
- **Tensorify.io:** Built a visual workflow compiler that transpiles node graphs into production-ready Python pipelines; executor service deployed on dedicated vCPU infrastructure with Linux-level sandboxing (`unshare`, `prlimit`) for multi-tenant workload isolation.
- **Flexx.dev:** Designed automated infrastructure provisioning and secure namespace isolation for a Redis-as-a-Service platform serving multi-tenant cloud customers.

Projects

Reactor — Autonomous AI Agent

LangGraph-based ReAct agent with file, shell, web, and API tools; stress-tested on a self-hosted webhook delivery platform.
[\[GitHub\]](#)

NodeBind11

High-performance C++ to Node.js binding generator enabling zero-copy data transfer and cross-language interoperability.
[\[GitHub\]](#)

Skills

- **Programming Languages:** Python, C++, Go, TypeScript, SQL
- **Machine Learning & AI:** PyTorch, LLMs, RAG Systems, LangChain, LangGraph, Computer Vision, 3D Point Clouds (PointNet++, ViT), Fine-Tuning, Vector Search
- **Distributed Systems & Infrastructure:** Redis, PostgreSQL, Docker, Kubernetes, Linux, Event-Driven Architectures, BullMQ, CQRS
- **Cloud Platforms:** AWS (EC2, S3), GCP, DigitalOcean, Hetzner
- **Software Engineering:** REST APIs, Microservices, CI/CD, Workflow Orchestration, Sandboxed Execution

References

Dr. Nabeel Mohammed

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Research Supervisor

Dr. Taseef H. Farook

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Research Collaborator, University of Adelaide

Saif Ahmed

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Research Supervisor